Report

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Explaination of aproch and design choices:

Backpropagation

$$\frac{\partial L}{\partial w_{i,j}^{[l]}} = \delta_i^{[l]} \, a_j^{[l-1]} \quad \forall l \in [1..L]$$

$$\delta_i^{[l]} = \frac{\partial L}{\partial z_i^{[l]}} = \frac{\partial L}{\partial a_i^{[l]}} \cdot \frac{\partial a_i^{[l]}}{\partial z_i^{[l]}}$$

$$\delta_i^{[L]} = \frac{\partial L}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial z_i^{[L]}} = e_i'(\hat{y}_i) \cdot f_i'^{[L]}(z_i^{[L]})$$

$$e_i'(y_i, \hat{y}_i) = -2(y_i - \hat{y}_i)$$

$$\delta_i^{[l]} = \left(\sum_{k=1}^{n^{[l+1]}} \delta_k^{[l+1]} w_{k,i}^{[l+1]} \right) \cdot f_i'^{[l]}(z_i^{[l]})$$

$$\frac{\partial L}{\partial b_j^{[l]}} = \delta_j^{[l]}$$

We used equation 4, 5, 6 & 7 to calculate our backpropagation. We decided that the best a

Gradiant descent: